



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2011 - 144



**3R Flat-Bed Grain Dryer
Model: 6T**

Test Report No.	:	2011 - 144
Date of Test	:	September 8, 2011
Test Valid Until	:	September 8, 2016
Brand	:	3R
Model	:	6T
Type	:	Batch, flat bed with automatic feeder rice hull-fed furnace
Manufacturer	:	3R METALCRAFT 40 Tandang Manang St., Parada Valenzuela City
Test Requested by:	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The 3R 6T Flat Bed Grain Dryer with an automatic feeder rice hull-fed furnace was tested for performance using PALAY as test material. The Dryer had a load capacity of 6 tons, however, only 5610 kg of palay was available during the time of test. It had a Drying Capacity of 510.0 kg/h from an initial grain Moisture Content of 26.8%_{w.b.} to a Final Moisture Content of 13.4%_{w.b.} with an average Moisture Reduction per Hour of 1.22%. The actual Drying Time for the palay with an initial weight of 5610 kg was 11.0 hours. The average Drying Air Temperature during the test was 49.8°C. The Fuel Consumption of the dryer's rice hull-fed furnace was 23.2 kg/h while the Fuel Consumption of the 9.2 kW KUVICO KV125-2 diesel engine which drove the hot air fan of the dryer was 0.80 L/h.

The Heating System Efficiency of the dryer was 70.7%. Heat Utilization was 2896 kJ per kg moisture removed with a corresponding Drying Efficiency of 82.1%. The dryer had a Drying System Efficiency of 58.1%. Based on the moisture content of samples taken at the end of drying (with manual mixing of grains during drying operation), Moisture Content Variation was $\pm 0.47\%$ while the Moisture Gradient from top to the bottom layer of the drying bin was 1.2%.

PURPOSE AND SCOPE OF TEST

The 3R 6T Flat Bed Grain Dryer was tested for drying PALAY to determine its operating performance. The test was conducted to verify and assess the following:

1.0 Drying System Performance

- 1.1 Drying Capacity
- 1.2 Moisture Reduction per Hour
- 1.3 Heat Utilization
- 1.4 Drying Efficiency
- 1.5 Drying System Efficiency

2.0 Heating System

- 2.1 Heating System Efficiency
- 2.2 Furnace Fuel Consumption Rate

3.0 Quality of Dried Grains

- 3.1 Cracked Grain
- 3.2 Milling Quality
- 3.3 Hulled/Damaged Grain
- 3.4 Moisture Content Variation
- 3.4 Moisture Content Gradient

PLACE AND DATE OF TEST

The 3R 6T Flat Bed Grain Dryer with an automatic feeder rice hull-fed type furnace was tested in Brgy. San Antonio, San Miguel, Iloilo on September 8, 2011.

METHODS OF TEST

The 3R 6T Flat Bed Grain Dryer was tested based on the Philippine Agricultural Engineering Standard “PAES 202:2000 Agricultural Machinery – Heated-Air Mechanical Grain Dryer - Methods of Test”.

OPERATOR’S MANUAL

No operator’s manual was submitted.

DESCRIPTION OF THE GRAIN DRYER

The 3R 6T Flat Bed Grain Dryer (**Figure 1**) was basically a non-recirculating batch type dryer with load capacity of 6 tons of palay. It consisted of the following main components: drying bin section, engine-driven fan and heating system.

Drying Bin (Figure 2)

The drying bin was a rectangular box which consisted of perforated floor and precast (panelized) concrete walls. The drying bin was designed for a volume capacity of approximately 6 tons at a grain layer thickness of 457 mm and provided with discharging chutes for bagging of dried grains.

Hot Air Fan (Figure 3)

The fan was of the tube axial type driven by a 9.2 kW KUVICO RV125-2 diesel engine (**Figure 4**) through a belt and a pulley transmission system. The fan has an overall diameter of 762 mm and draws hot air from a rice hull-fed furnace through a metal sheet transition duct, pushes it to the plenum chamber and through the grain mass in the drying bin.

HEATING SYSTEM

The heating system employed a direct-fired rice hull-fed furnace with automatic feeding mechanism. It consisted of the following components: hopper, furnace, and fan for supply of combustion air.

Hopper (Figure 5a)

This was where fuel was received. It was provided with an auger (**Figure 5b**) which regulated the amount of fuel being fed to the combustion chamber. The auger was also driven by the KUVICO KV125-2 diesel engine through a belt and pulley transmission system, a speed reducer (**Figure 6**) and through a chain and sprocket transmission system.

Furnace (Figure 7a)

This was made of fire bricks where combustion took place. Fuel was fed from the auger of the hopper and at the opposite end was the combustion chamber

(**Figure 7b**) with inclined flooring with perforations for entry of combustion air. Ash was discharged at the end of the inclined flooring and manually collected.

Fan for Supply of Combustion Air (**Figure 8**)

It was a centrifugal fan which provided primary air underneath the combustion chamber through holes in the bricks floor. It was also driven by the 9.2 kW KUVICO KV125-2 diesel engine through belt and pulley transmission system.



Figure 1. 3R Flat Bed Grain Dryer
Model: 6T



Figure 2. Drying Bin

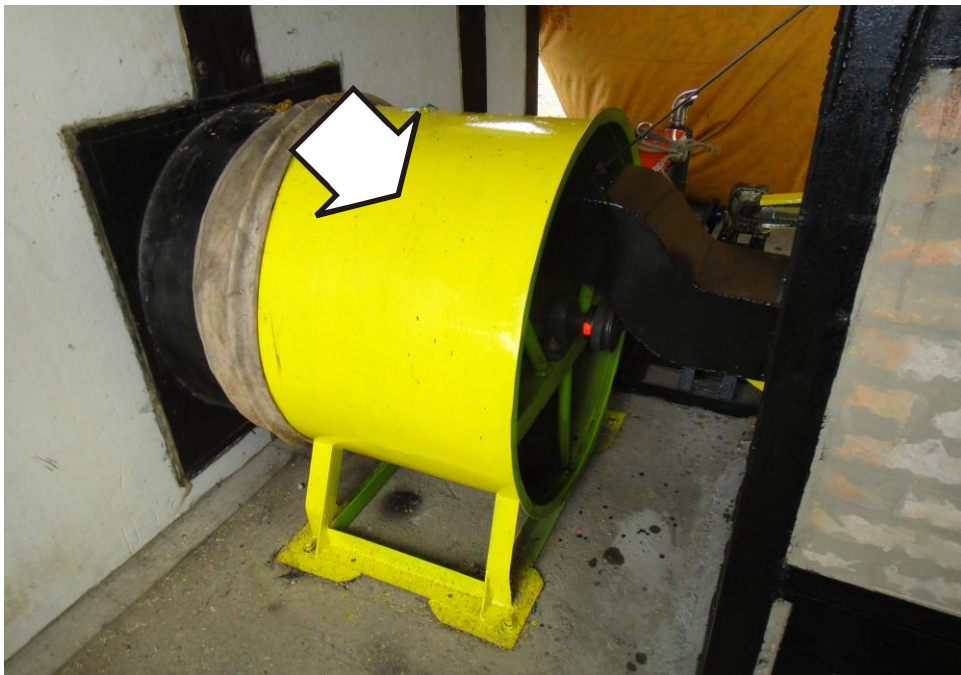


Figure 3. Hot Air Fan



Figure 4. Dryer's Prime Mover

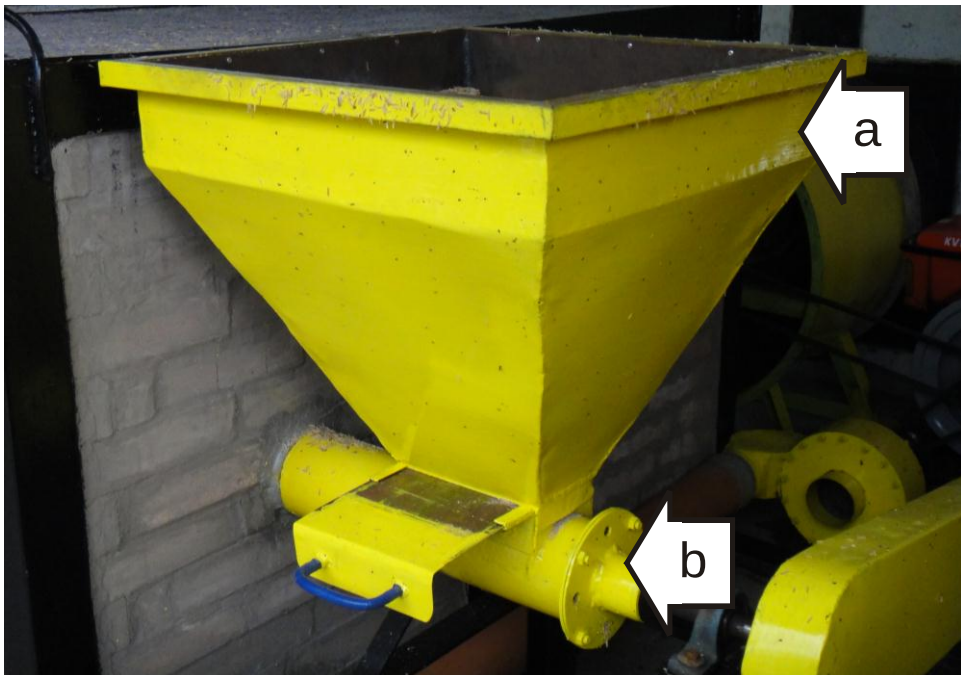


Figure 5. a) Hopper
b) Auger

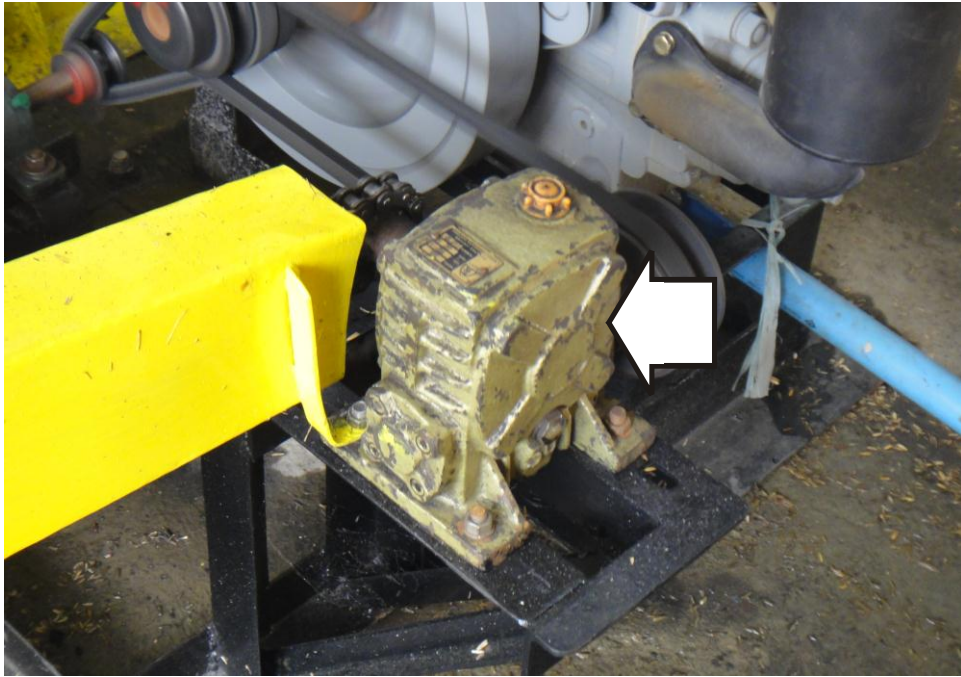


Figure 6. Speed Reducer

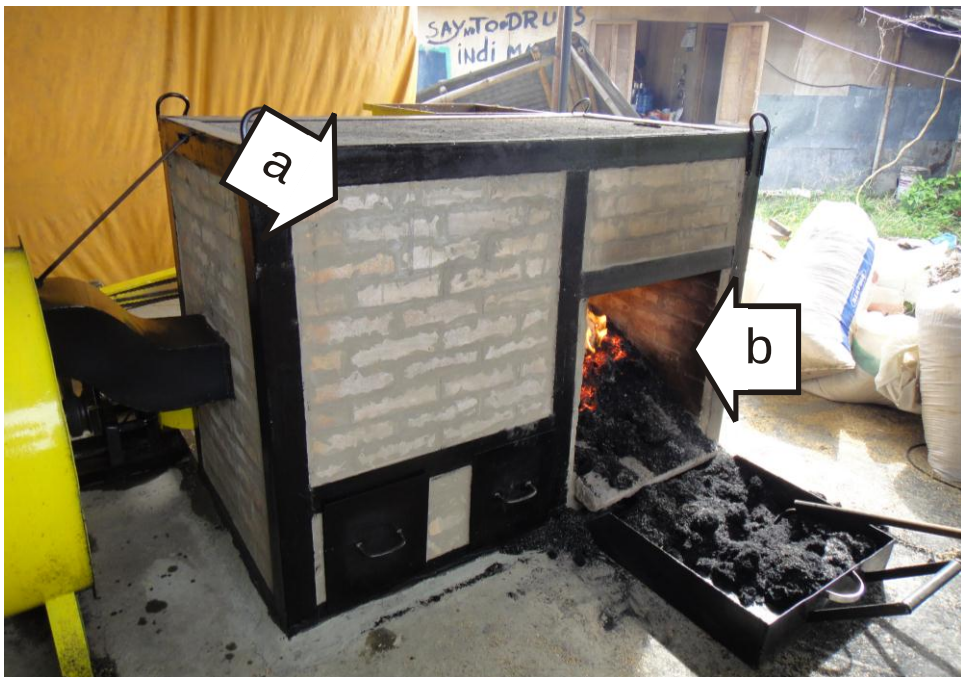


Figure 7. a) Furnace
b) Combustion Chamber



Figure 8. Fan for Supply of Combustion Air

RESULTS AND DISCUSSION

The 3R 6T Flat Bed Dryer with a 6-ton holding capacity was tested by filling up the drying bin with 5610 kg of fresh PALAY. The furnace was fired and the feeding rate of rice hull was set to control the drying air temperature. Manual mixing of grains was done twice during the drying operation. Drying operation was stopped when the average moisture content of the grains from top to bottom layer of the drying bin was about 14.0%.

Drying Capacity

The drying capacity of the dryer is the amount of fresh palay dried per unit of time, from initial moisture content until the desired moisture content is attained. The dryer had a Drying Capacity of 510.0 kg/h from an initial moisture content of 26.8%_{W.B.} to a final moisture content of 13.4%_{W.B.}.

Moisture Content Reduction

The average Moisture Content Reduction during the drying process was 1.22% per hour.

Moisture Content Variation and Moisture Gradient

Based on the moisture content of samples randomly taken from the drying bin, Moisture Content Variation at the termination of drying was $\pm 0.47\%$ with Moisture Gradient of 1.2% (with manual mixing of grains during the drying operation).

Heating System Efficiency

Heating system efficiency is the ratio of the amount of heat supplied for drying to the total amount of heat available from the fuel, expressed in percent. The dryer had Heating System Efficiency of 70.7%.

Heat Utilization

Heat utilization refers to the total amount of heat utilized to vaporize moisture from the grain. Results showed that the dryer had a Heat Utilization of 2896 kJ/kg of water removed.

Drying Efficiency

Drying efficiency is the ratio of the total heat utilized to vaporize moisture in the grain to the amount of heat added to the drying air expressed in percent. Results showed that the dryer had a Drying Efficiency of 82.1%

Drying System Efficiency

Drying system efficiency is the ratio of the total heat utilized for drying to the heat available in the fuel expressed in percent. Results showed that the dryer had a Drying System Efficiency of 58.1%.

Furnace's Fuel Consumption

The fuel consumption of the rice hull-fed furnace was determined by getting the total amount of rice hull consumed during the test divided by the total time of operation. The Fuel Consumption of the furnace was 23.2 kg/h of rice hull.

Engine's Fuel Consumption

The Fuel Consumption of the 9.2 kW KUVICO KV125-2 diesel engine which drove the hot air fan and the fan for combustion air was 0.80 L/h.

Labor Requirements

Two persons were needed for loading, mixing and unloading of grains while one person was needed in operating the dryer (**Figure 9**).

Observations

The dryer operated smoothly and no breakdown or failure of components occurred during the test.

The following were also observed:

1. The dryer was provided with a dial type thermometer installed at the drying bin's plenum (**Figure 10**).
2. The dryer's transmission system was provided with belt cover (**Figure 11**)
3. Manual mixing of grains (**Figure 12**) was conducted to ensure uniform grain moisture content at the termination of drying operation.



Figure 9. Operator of the dryer



Figure 10. Dial thermometer

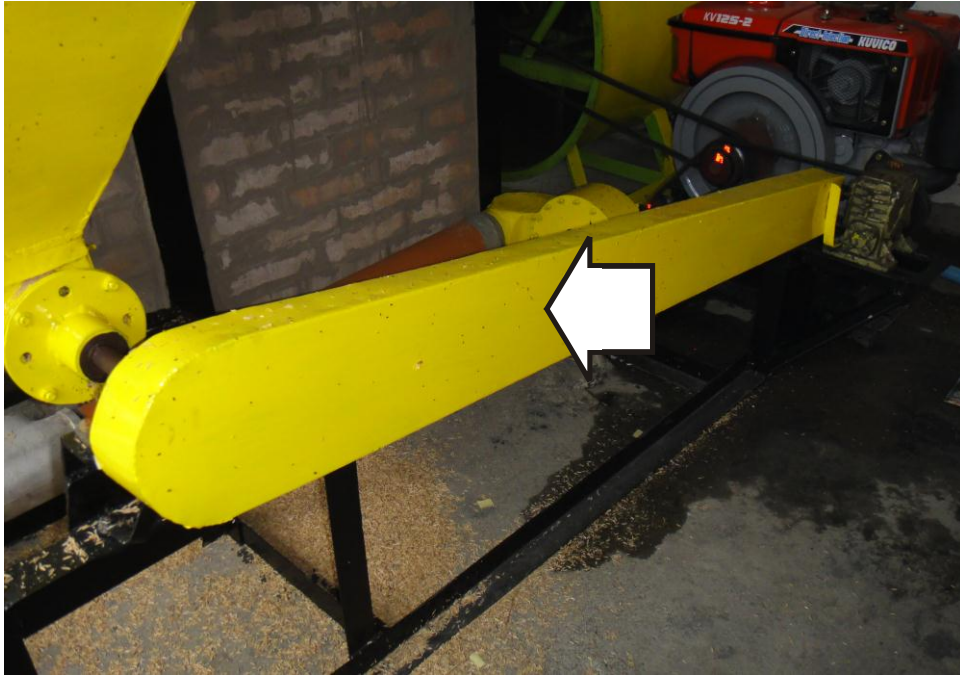


Figure 11. Transmission System Cover



Figure 12. Manual mixing of grains

Table 2. Results of Performance Test of 3R 6T Flat Bed Grain Dryer

ITEMS	DATA
1. Initial Weight of Test Material (kg)	5610
2. Final Weight of Test Material (kg)	4742
3. Initial Moisture Content (%)	26.8
4. Final Moisture Content (%)	13.4
5. Actual Drying Time (h)	11.0
6. Drying Capacity (kg/h)	510
7. Rate of Moisture Removal (kg/h)	78.9
8. Moisture Reduction Rate (%/h)	1.22
9. Average Drying Air Temperature (°C)	49.8
10. Ambient Air Temperature (°C)	
a) Dry Bulb	30.0
b) Wet Bulb	27.4
11. Ambient Air Relative Humidity (%)	81.9
12. Dryer Exhaust Air Temperature (°C)	
a) Dry Bulb	32.8
b) Wet Bulb	31.4
13. Exhaust Air Relative Humidity (%)	90.5
14. Grain Temperature (°C)	34.6
15. Fan Air Velocity (m/s)	5.94
16. Air Flow Rate (m ³ /min)	163.8
17. Plenum Static Pressure (mmH ₂ O)	8
18. Speed of Components (rpm)	
a) Engine	1621
b) Fan Shaft	1001
c) Auxiliary Fan Shaft	3357
d) Rice Hull Screw Conveyor Shaft	10.4
19. Rice Hull Consumption Rate (kg/h)	23.2
20. Engine's Fuel Consumption Rate (L/h)	0.80
21. Heating System Efficiency (%)	70.7
22. Heat Utilization (kJ/kg moisture removed)	2896
23. Drying Efficiency (%)	82.1
24. Drying System Efficiency (%)	58.1
25. Moisture Content Variation at the Termination of Drying (%)	±0.47
26. Moisture Gradient (%)	1.2

Table 3. General information of air-dried palay samples brought to AMTEC laboratory for analysis

ITEMS	DATA
1. Variety	RC-10
2. Source	San Miguel, Iloilo
3. Paddy Analysis	
3.1 Grain Size, brown rice (mm)	
3.1.1 Length	6.80
3.1.2 Width	2.30
3.1.3 Thickness	1.71
3.2 Grain Type	Long and slender
3.3 Paddy Bulk Density (kg/m ³)	528.8
3.4 Moisture Content (% w. b.)	14.0
3.5 Purity (%)	94.80
3.6 Cracked (%)	0
3.7 Immature (%)	0.40
3.8 Chalky (%)	14.6
3.9 Yellow and Fermented (%)	12.1
3.10 Red Rice (%)	2.32
4. Laboratory Milling Test	
4.1 Total Milling Recovery (%)	67.9
4.2 Head Rice Recovery (%)	
4.2.1 Based on Milled Rice	79.7
4.2.2 Based on Rough Rice	54.1
4.3 Broken Rice (%)	
4.3.1 Based on Milled Rice	18.9
4.4 Brewer's Rice	
4.4.1 Based on Milled Rice	1.4

Table 4. General information of samples dried using the 3R 6T Flat Bed Grain Dryer brought to AMTEC laboratory for analysis

ITEMS	DATA
1. Variety	RC-10
2. Source	San Miguel, Iloilo
3. Paddy Analysis	
3.1 Grain Size, brown rice (mm)	
3.1.1 Length	6.80
3.1.2 Width	2.27
3.1.3 Thickness	1.76
3.2 Grain Type	Long and slender
3.3 Paddy Bulk Density (kg/m ³)	545.0
3.4 Moisture Content (% w. b.)	13.4
3.5 Purity (%)	95.38
3.6 Cracked (%)	20.0
3.7 Immature (%)	0.52
3.8 Chalky (%)	15.7
3.9 Yellow and Fermented (%)	11.8
3.10 Red Rice (%)	0
4. Laboratory Milling Test	
4.1 Total Milling Recovery (%)	67.6
4.2 Head Rice Recovery (%)	
4.2.1 Based on Milled Rice	77.0
4.2.2 Based on Rough Rice	52.1
4.3 Broken Rice (%)	
4.3.1 Based on Milled Rice	21.0
4.4 Brewer's Rice	
4.4.1 Based on Milled Rice	2.0

ANNEX 1. FORMULA USED IN THE COMPUTATION OF DRYING PARAMETERS

$$1. \quad \text{Drying Capacity (kg/h)} = \frac{W_1}{T_A}$$

where,

W_1 = Initial weight of test material (kg)

T_A = Actual drying time (h)

$$2. \quad \text{Final Weight of Test Materials, (kg)}$$

$$W_2 = \frac{W_1 (100 - MC_1)}{(100 - MC_2)}$$

Where,

W_2 = Final weight of test materials (kg)

MC_1 = Initial moisture content (%)

MC_2 = Final moisture content (%)

$$3. \quad \text{Moisture Reduction Per Hour (kg/h)}$$

$$\text{Moisture reduction per hour} = \frac{W_1 - W_2}{T_A}$$

$$4. \quad \text{Heating System Efficiency, } E_{HS} \text{ (%)}$$

$$E_{HS} = \frac{Q_s}{Q_f} \times 100$$

Where,

E_{HS} = Heating system efficiency (%)

Q_s = Heat supplied to the dryer (kJ/h)

Q_f = Heat available in the fuel (kJ/h)

To solve for Q_s :

$$Q_s = \frac{\Delta h \times Q}{v} \times 60$$

Where,

Δh = Change in enthalpy (kJ/kg d.a.)

Q = Air flow rate (m³/min)

v = Specific volume of fresh air (m³/kg d.a.)

To solve for Q_f :

$$Q_f = F_c \times H_{vf}$$

Where,

F_c = Fuel consumption (kg/h)

H_{vf} = Heating value of the fuel used (kJ/kg)

5. Heat utilization, Q_u (kJ/kg)

$$Q_u = \frac{Q_s \times T_A}{W_1 - W_2}$$

6. Drying Efficiency (%)

$$E_D = \frac{Q_R}{Q_S} \times 100$$

Where,

E_D = Drying efficiency (%)

Q_R = Heat required to vaporize moisture inside the grain (kJ/h)

To solve for Q_R

$$Q_R = Q_v - (W_1 - W_2)/T_A$$

Where,

Q_v = Heat of vaporization of water

7. Drying System Efficiency, E_{DS} (%)

$$E_{DS} = Q_T/Q_f \times 100$$

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

This test report should not be used for advertisement or for promotion of the product but only for evaluation purposes and should not be reproduced in whole or in part without the permission of AMTEC.